

Explainability

Machine Learning for Behavioral Data

April 21, 2026

Vinitra Swamy

Today's Topic

Week	Lecture/Lab
7	Unsupervised Learning
8	Spring Break
9	Unsupervised Learning
10	Explainability
11	Fairness
12, 13, 14	Work on your projects
15	Poster Presentations



- What is fairness?
- Fairness metrics
- Interpreting neural networks

Getting ready for today's lecture...

- **If not done yet:** clone the repository containing the Jupyter notebook and data for today's lecture into your Noto workspace
- SpeakUp room for today's lecture:

[https://web.speakup.info/room/join/
56580](https://web.speakup.info/room/join/56580)



Short quiz about the past...

In K-Means Clustering, how should you initialize the cluster centroids?

- a) Once, randomly
- b) Once, uniformly
- c) Visualizing the data and picking appropriate starting points
- d) Multiple times randomly and minimizing distortion

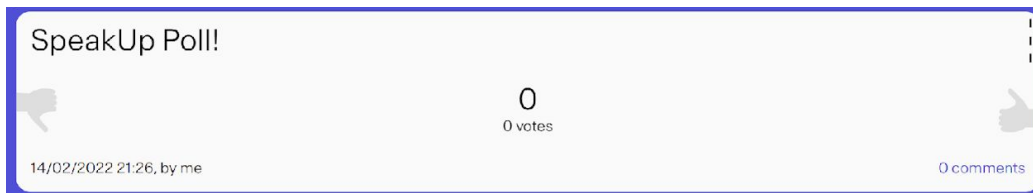


<https://web.speakup.info/room/join/56580>

Short quiz about the past...

When performing clustering on text data, which distance/similarity metric is appropriate?

- a) Silhouette Score
- b) Jaccard Similarity
- c) Cosine Similarity
- d) Euclidean Distance



<https://web.speakup.info/room/join/56580>

Short quiz about the past...

If you use accuracy instead of balanced accuracy for a binary classification task on an imbalanced data set, this is an example of:

- a) Historic Bias
- b) Evaluation Bias
- c) Measurement Bias
- d) Aggregation Bias



<https://web.speakup.info/room/join/56580>

Short quiz about the past...

You are building a model for whether someone will pass a class based on their MOOC clickstream. You are concerned about whether your model's predictions of passing and predictions of failing are equally accurate across demographic groups. Which metric do you use?

- a) equalized odds
- b) demographic parity
- c) predictive (value) parity



<https://web.speakup.info/room/join/56580>

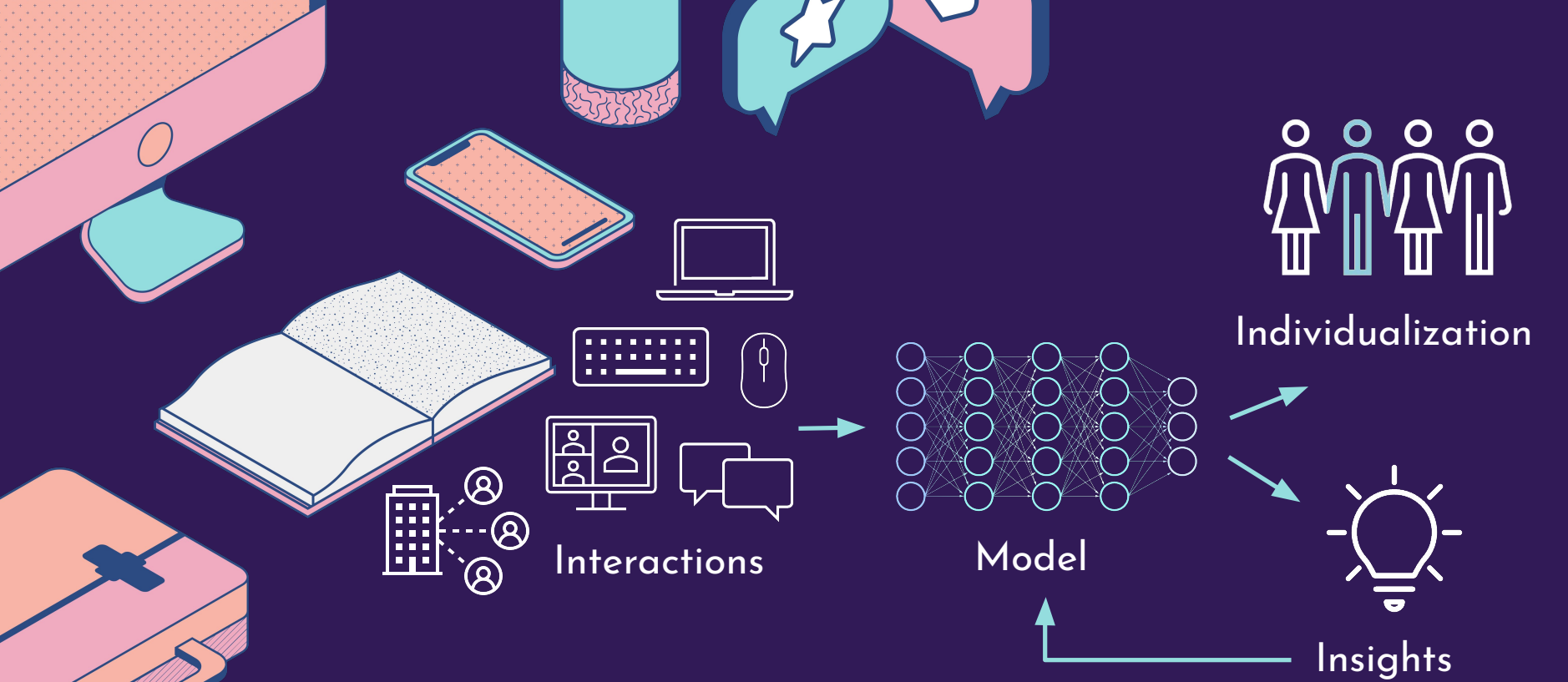
Agenda

1) Introduction to Explainability

- Taxonomy of interpretability methods
- Deep Dive: LIME
- Deep Dive: PDP

2) Personalized Learning with Scholé!

- Explainable AI
-



**Student models enable
individualization and insights**

Learning Objectives

You should be able to:

- Describe and categorize the explainability methods discussed in class
 - Explain their strength and weaknesses
 - Interpret their outputs
 - Apply the methods (using the APIs) to predictions of a model and discuss the results
-

Interpretability

Interpretability is the degree to which a human can understand the cause of a decision.

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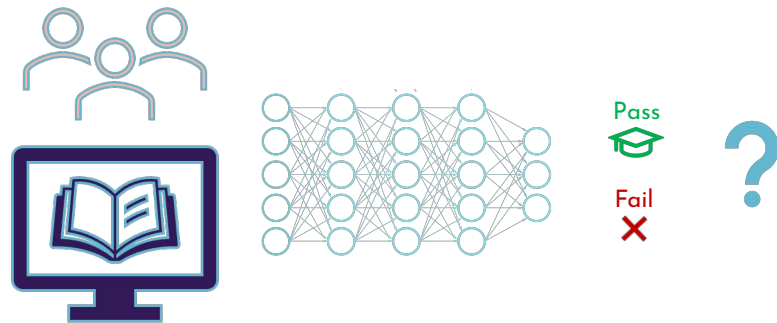


The higher the interpretability of a machine learning model, the easier it is for someone to comprehend why certain predictions have been made

Cost of using neural networks

DEEP LEARNING IN
EDUCATION

Problem: Deep Learning trades
transparency for accuracy



Identifying “why” is important for effective, personalized interventions

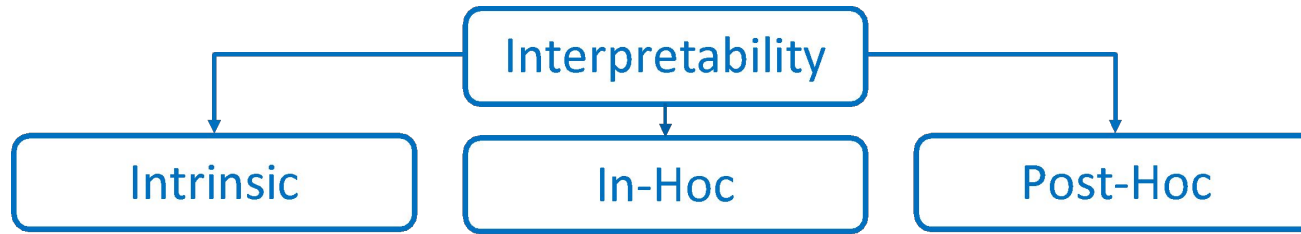
Solution: Explainable Machine Learning

Why is eXplainable AI important?

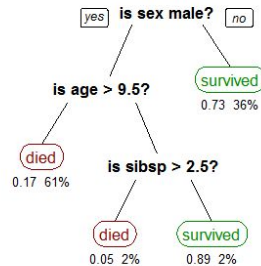
1. Building student / teacher trust in models
2. Auditing models when they make mistakes
3. Improving models with their own reasoning

Interpretability

XAI Fundamentals



Decision Trees

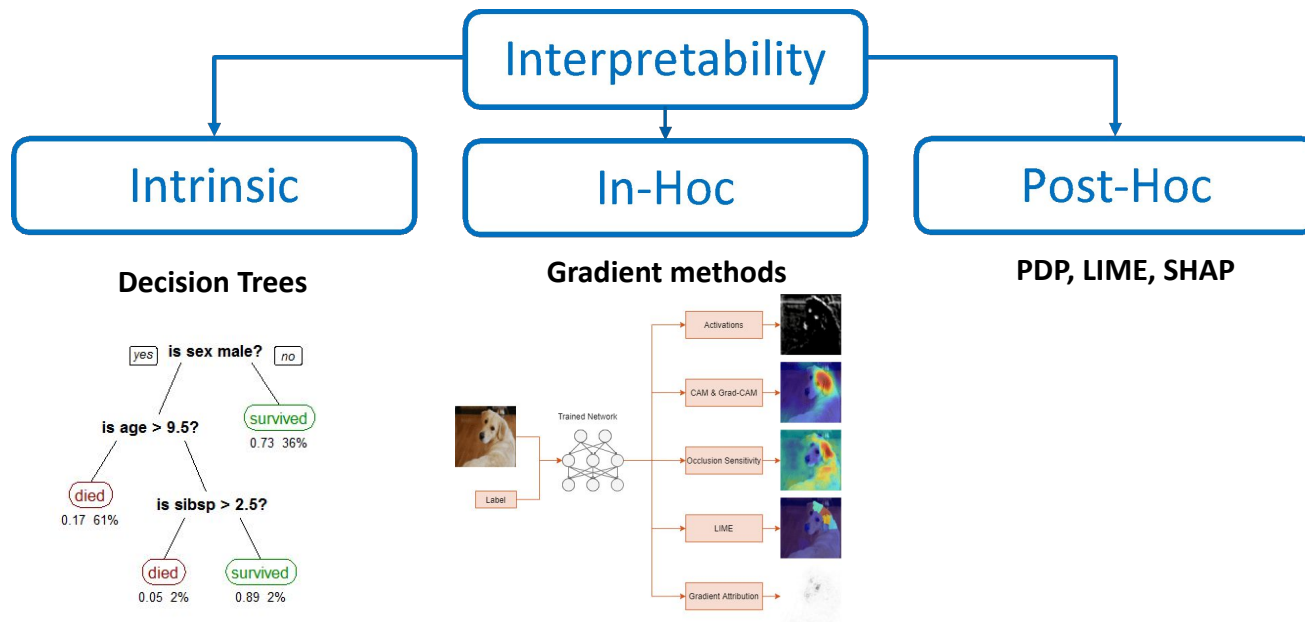


Gradient methods

PDP, LIME, SHAP

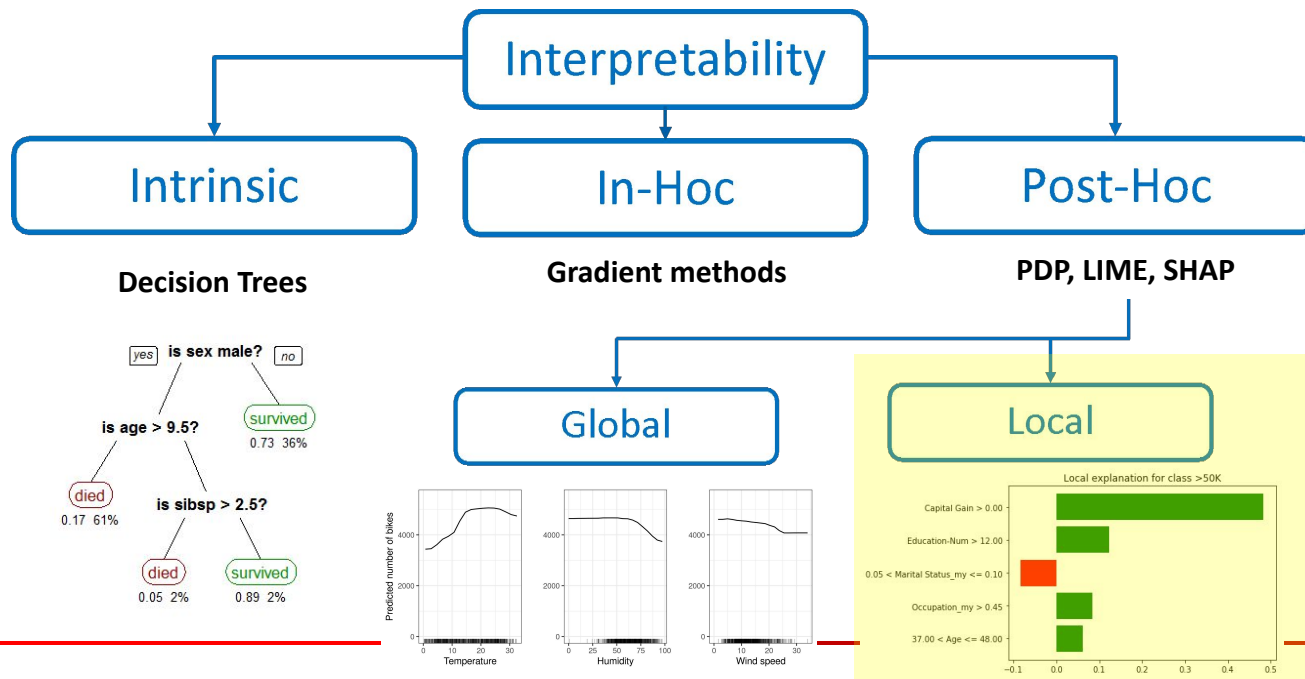
Interpretability

XAI Fundamentals



Interpretability

XAI Fundamentals



LIME

Local Interpretable Model-Agnostic Explanations

1

Select a specific point to explain: $(X_{\text{student}}, Y_{\text{student}})$



X_{student}

2

Perturb features of selected point to get $\{X^1_{\text{student}} \dots X^N_{\text{student}}\}$ neighbors

Black-Box Model



Fail
X

Y_{student}

3

Feed in X_{student} neighbors to the black-box model and get predictions $\{Y^1_{\text{student}} \dots Y^N_{\text{student}}\}$

Pass
X

Fail
X

Pass
X

Pass
X

Fail
X

Fail
X

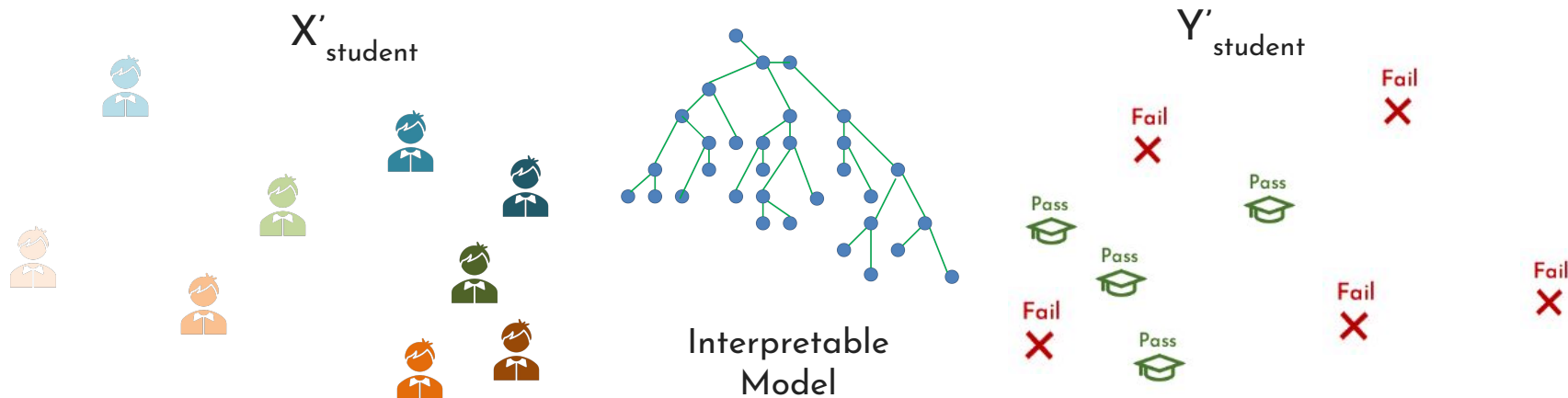
Fail
X

LIME

Local Interpretable Model-Agnostic Explanations

1
9
4

Train an interpretable local model using (weighted) X'_{student} and Y'_{student}



Local interpretable model-agnostic explanations (LIME)

- Idea: use a local surrogate model (interpretable) to explain individual predictions of a black-box model

$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

LIME - Recipe

$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

1. Select your instance (sample) of interest
-

LIME - Recipe

$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

2. Perturb your data set: generate new samples that are variations of the selected sample
-

LIME - Recipe

$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

3. Get the black-box model predictions for the new samples
-

LIME - Recipe

$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

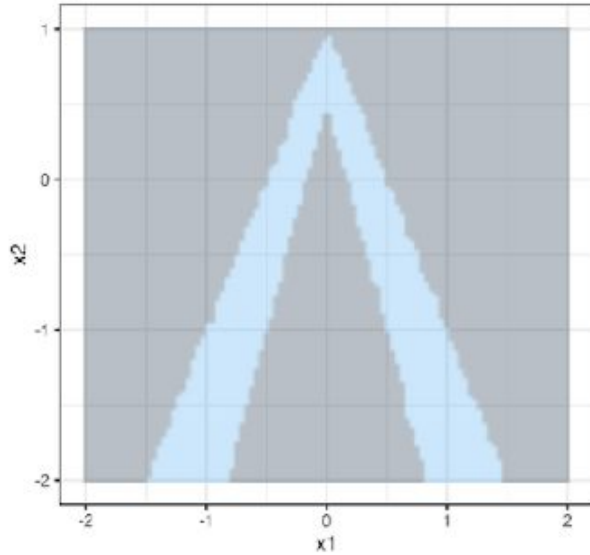
4. Train a weighted, interpretable model on the data set with variations
-

LIME - Recipe

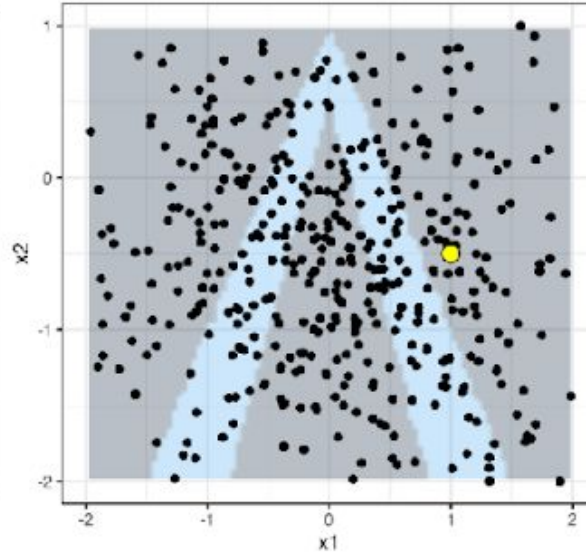
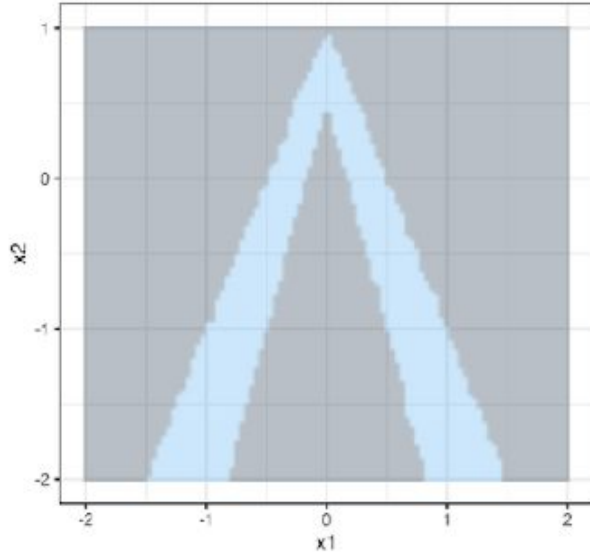
$$\text{explanation}(x) = \underset{g \in G}{\operatorname{argmin}} L(f, g, \pi_x) + \Omega(g)$$

5. Explain the prediction by interpreting the local model
-

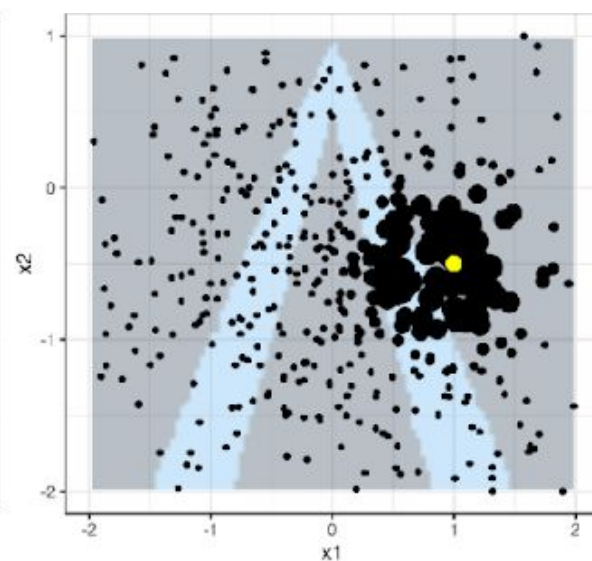
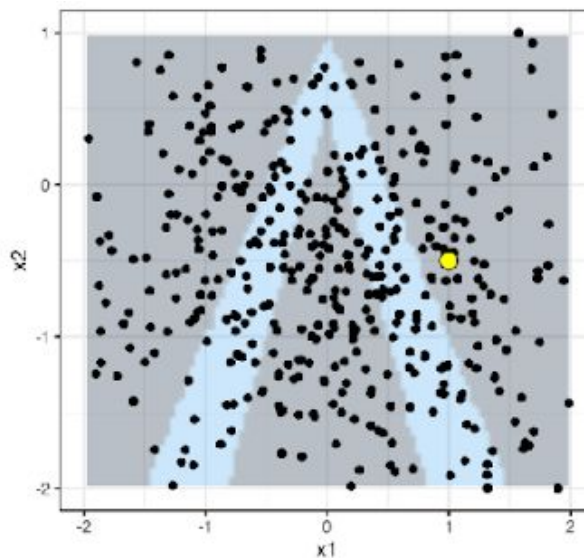
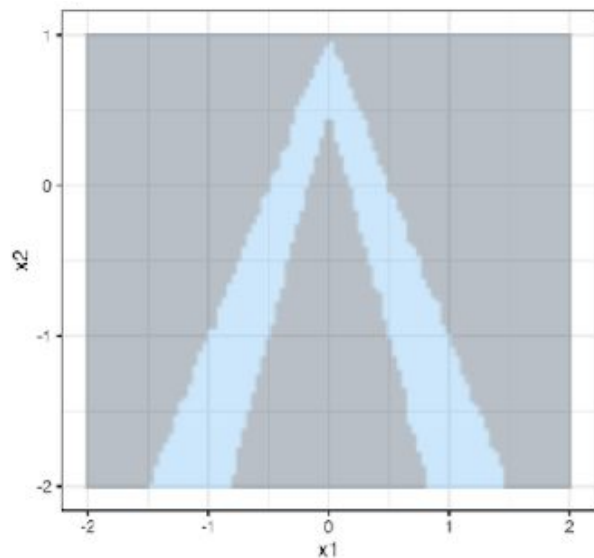
LIME – Perturbation of Sample



LIME – Perturbation of Sample



LIME – Perturbation of Sample

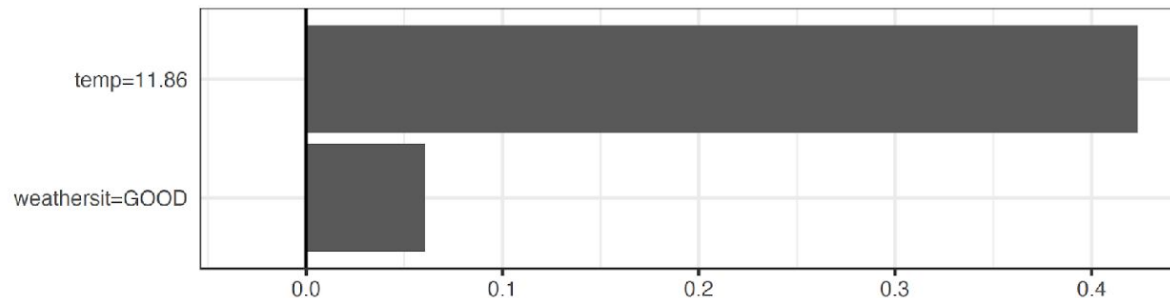


Example – Bike Rental Shop

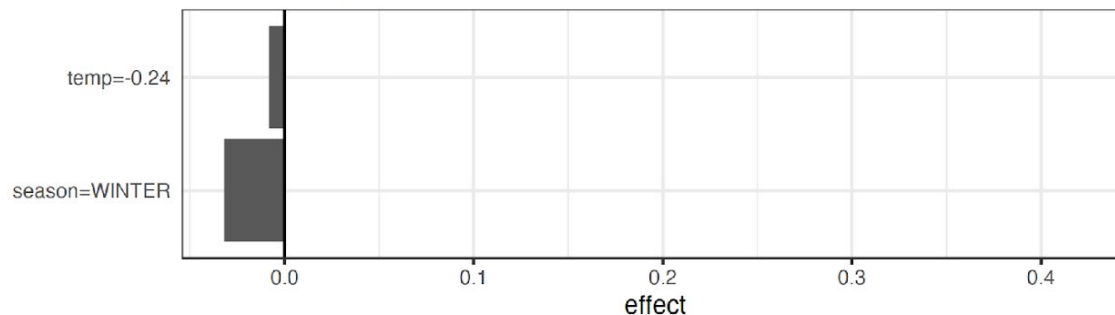
- Y is binary and indicates, whether the number of bikes rented on a given day will be **above average** ($y = 1$)

Example – Bike Rental Shop

Actual prediction: 0.89
LocalModel prediction: 0.44



Actual prediction: 0.01
LocalModel prediction: -0.03



LIME – Strengths and Weaknesses

- + Model-agnostic (we can replace the underlying model and still use the same surrogate model)
 - + When using for example Lasso regression, explanations are short (= selective)
 - + Benefit from literature on training and interpreting interpretable models
 - + Fidelity measure gives us an idea of reliability
 - Definition of local neighborhood unsolved problem
 - Sampling ignores correlation between features (-> unlikely data points)
 - Instability of explanations
-

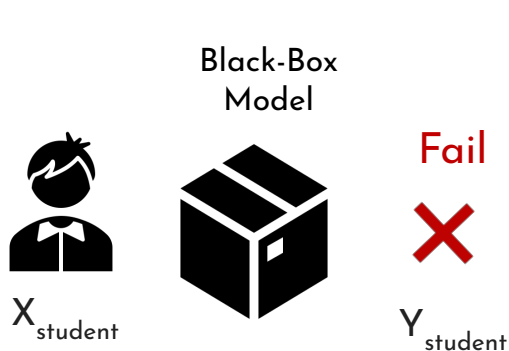
LIME – Your Turn

- Your Task:
 1. Run LIME on two instances of your choice
 2. Share the plots for the two instances with us as well as your observations (Are the same features important for both instances? Can you interpret the feature effects?)
-

SHAP

SHapley Additive exPlanations

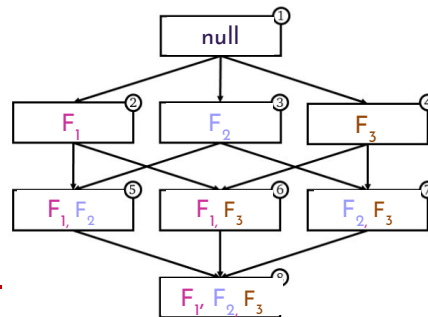
SHAP explains X_{student} by quantifying the contribution of each feature to the prediction.



F_1 F_2 F_3

{ # of minutes watching videos , # of clicks on problems this week , # of sessions (overall) }

Power Set
(coalition)



cardinality

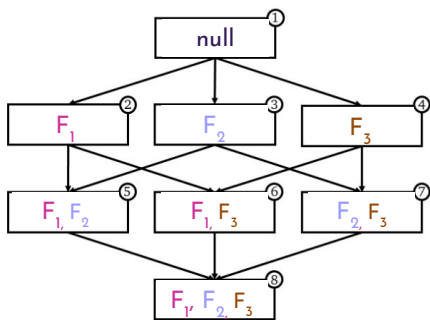
2^3

SHAP

SHapley Additive exPlanations

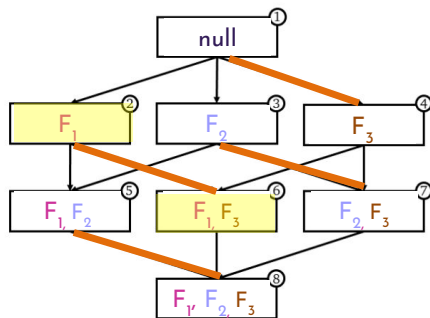
1

Train a model on each feature coalition.



2

Weighted sum of “marginal contributions” for each feature (i.e. F_3).



KernelSHAP

Optimizations using the **SHAP kernel function** for efficient data point construction

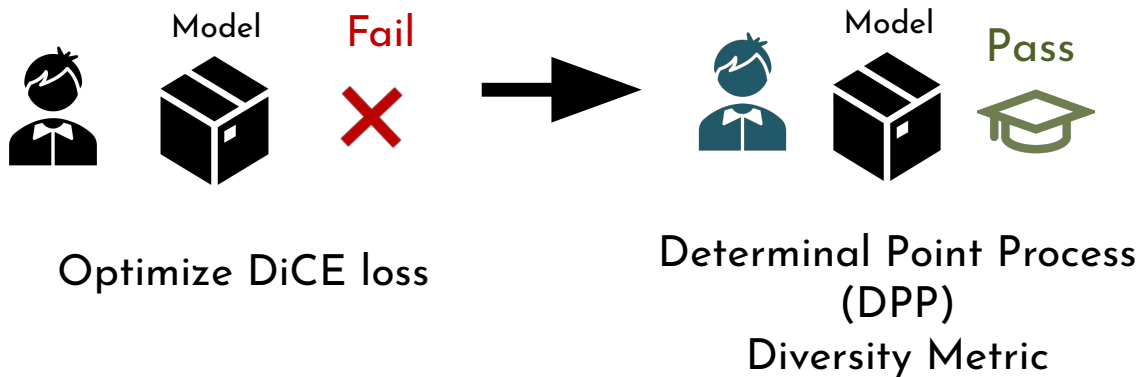
PermutationSHAP

All feature combinations in forward and reverse directions
(antithetic sampling)

DiCE

Diverse Counterfactual Explanations for ML

Generate a point with the smallest possible change to the initial instance that results in a different prediction.



CEM

Contrastive Explanation Method

$$\{ F_1, F_2, F_3, F_4, \dots, F_{42}, F_{43}, F_{44}, F_{45} \}$$

Pertinent Positives (PP)

X' with the minimal subset of features that should be **present** to maintain the prediction.

Pertinent Negatives (PN)

X' with a subset of features **absent** while maintaining the prediction.

$$\text{Feature importance: } |X'_{\text{student_k}} - X_{\text{student_k}}| \times SD_{\text{feature}}$$

Global Method: Partial Dependency Plot (PDP)

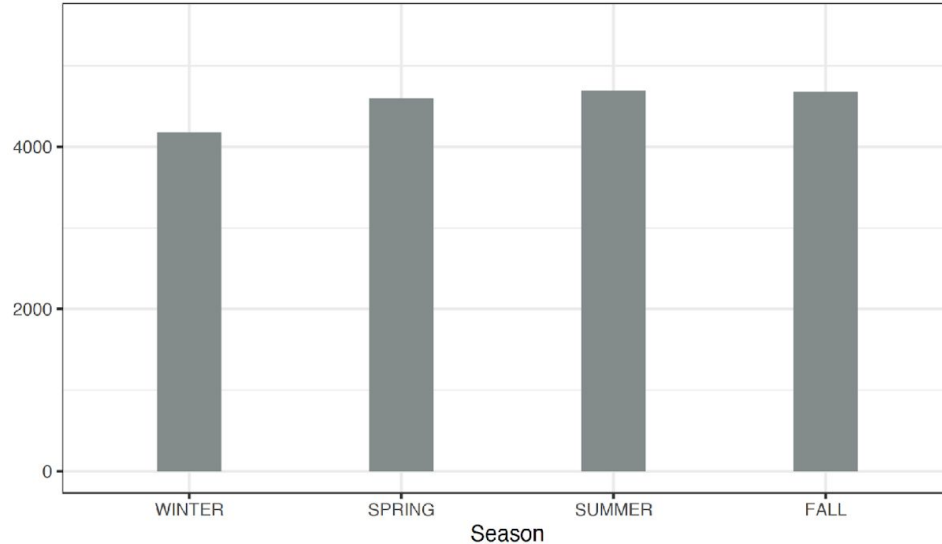
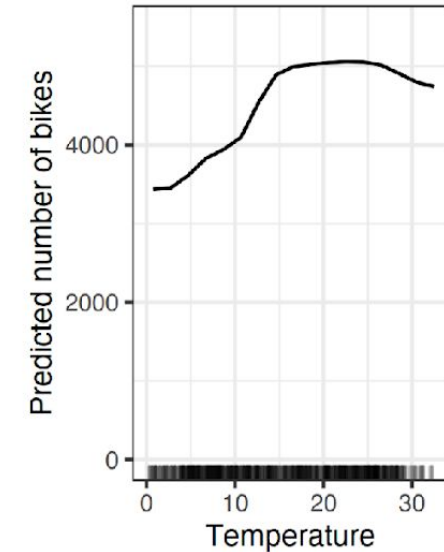
- PDP is model-agnostic
 - PDP show the marginal effects a subset of features have on the predicted outcome of a model
 - The subset of features usually consists of one feature (resulting in a 2D-Plot) or two features (resulting in a 3D-Plot)
-

Example – Bike Rental Shop

- Y denotes the number of bikes that will be rented on a given day
- Features (X): season, work day, temperature, humidity, ...

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Partial Function - Regression

$$\hat{f}_S(x_S) = E_{X_C}[\hat{f}_S(x_S, X_C)] = \int_{X_C} \hat{f}_S(x_S, X_C)$$

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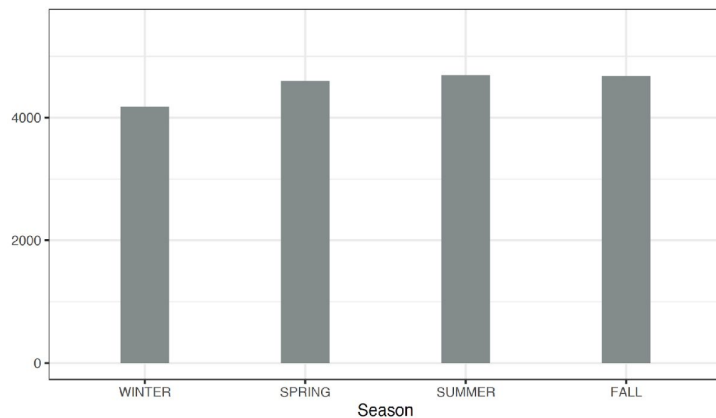
$$\hat{f}_S(x_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}_S(x_S, x_c^{(i)})$$

Partial Function - Regression

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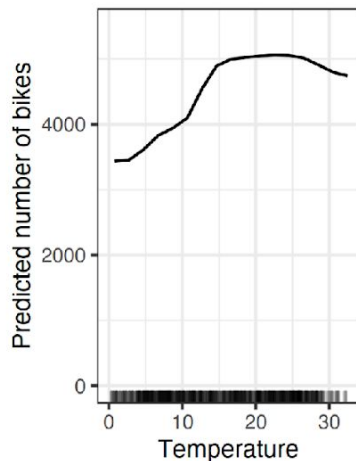
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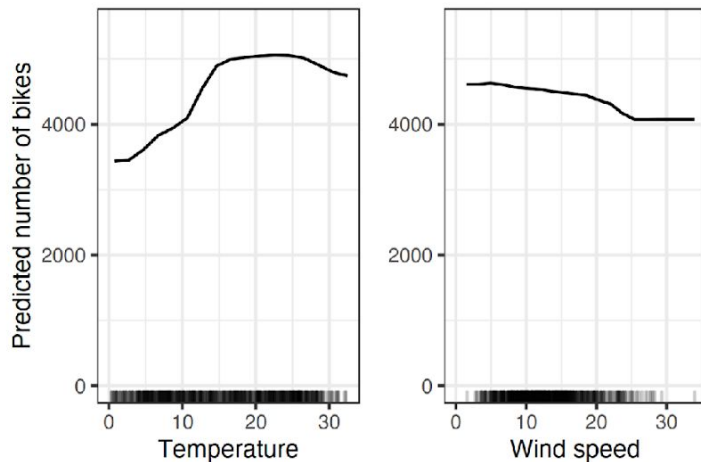
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Partial Function - Regression

$$\hat{f}_S(x_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}_S(x_S, x_c^{(i)})$$



Partial Function - Classification

$$\hat{f}_S(x_S) = \frac{1}{n} \sum_{i=1}^n \hat{f}_S(x_S, x_c^{(i)})$$

- If classifier outputs a probability, the PDP displays the probability for a certain class given different values for feature(s) in S
 - Dealing with multiple classes: draw one line or plot per class
-

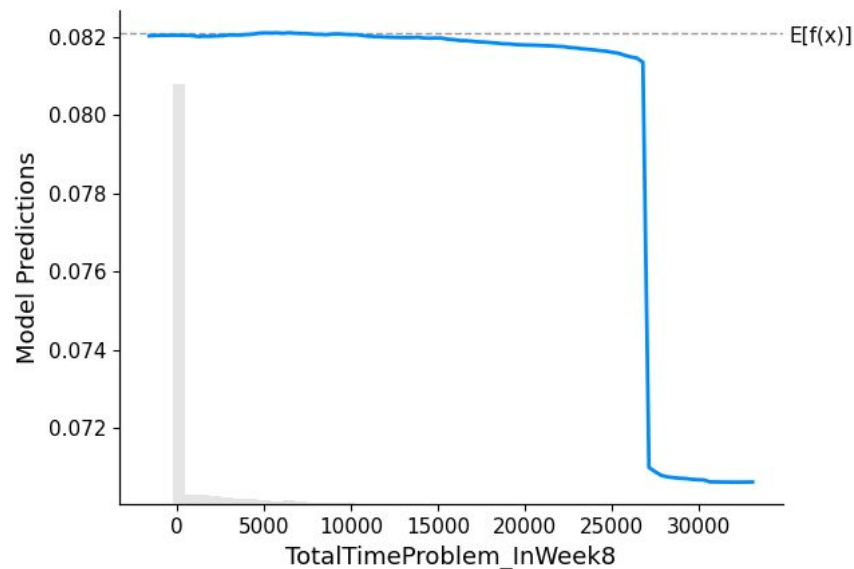
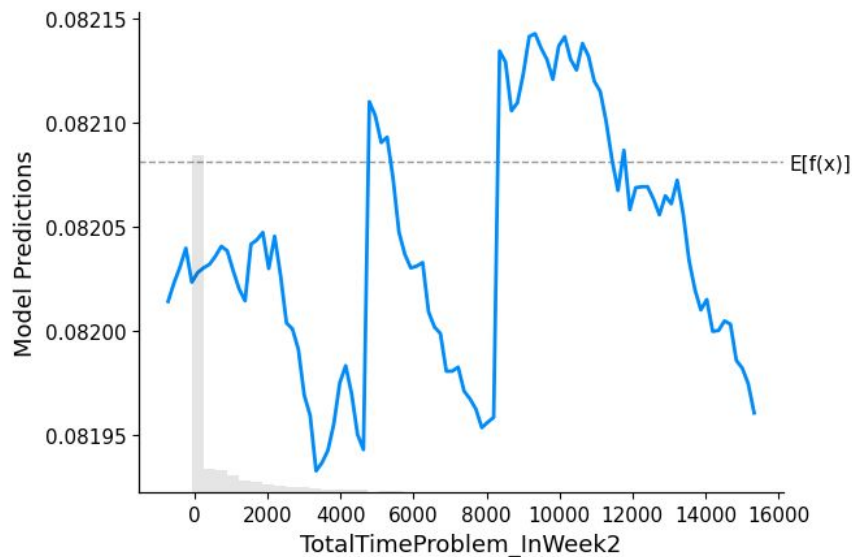
PDP – Strength & Weaknesses

- + Model-agnostic
 - + Computation is intuitive, interpretation is clear
 - + Easy to implement
 - + Causal interpretation
 - Maximum number of features in a PDP is two
 - Assumption of independence
 - Some PDP do not show feature distribution
-

PDP – Your Turn

- Participants: 8679 students of a of an EPFL MOOC with a duration of 10 weeks
 - We have trained a classifier to predict whether a student will pass or fail the course based on their clickstream data
 - Your Task:
 1. Investigate the PDPs for *TotalTimeProblem* in week 2 and week 8
 2. Discuss: how does this feature influence predictions? Is there a difference between week 2 and week 8? What about the distribution of feature values?
-

PDP Example – EPFL MOOC





An Outlook on Recent Research

1. Interpretability is **crucial** in human-centric AI
 2. Current post-hoc explainability methods have flaws
 3. **Inherently interpretable** neural network architectures show promise
-

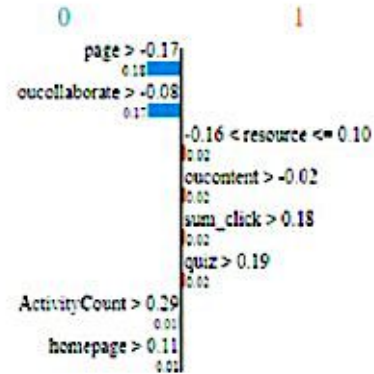
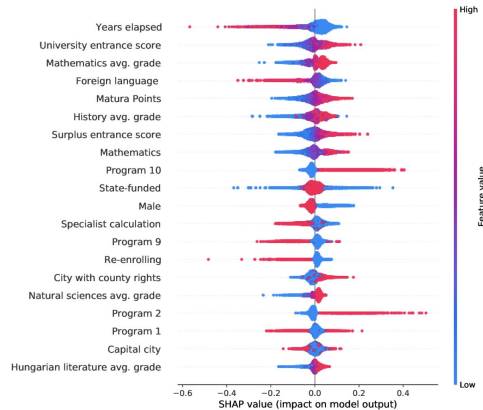
Previous Work

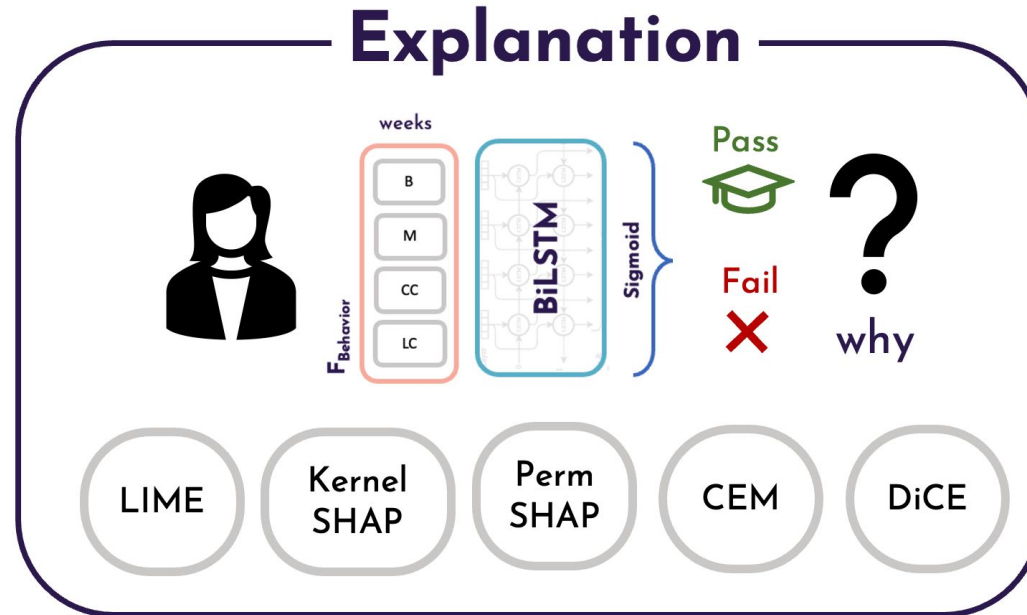
MOTIVATION

Previous work: In (minimal) related literature, only one explainability method is picked per application

SHAP for student dropout^[1]

LIME for student advising^[2,3]





Explanation: How important is this feature to the model's prediction?

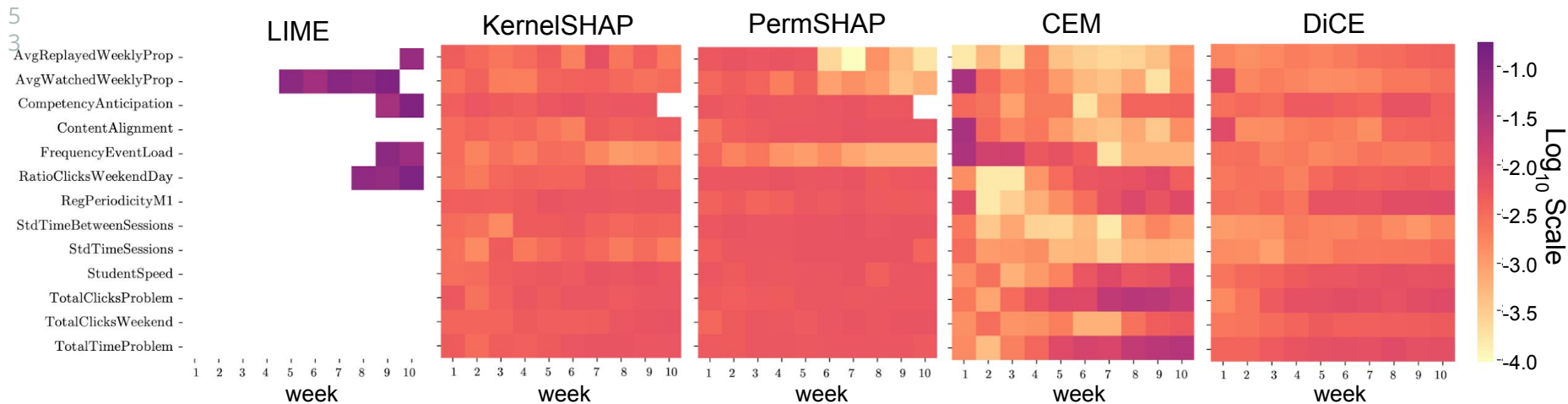
RQ1: 1 Course

RESULTS



[EDM 2022] Evaluating the Explainers
Swamy, Radmehr, Krco, Marras, Käser

How similar are the explanations of different explainability methods for a specific course (DSP 1)?



LIME is very sparse. CEM is significantly different.

RQ2: 5 Courses

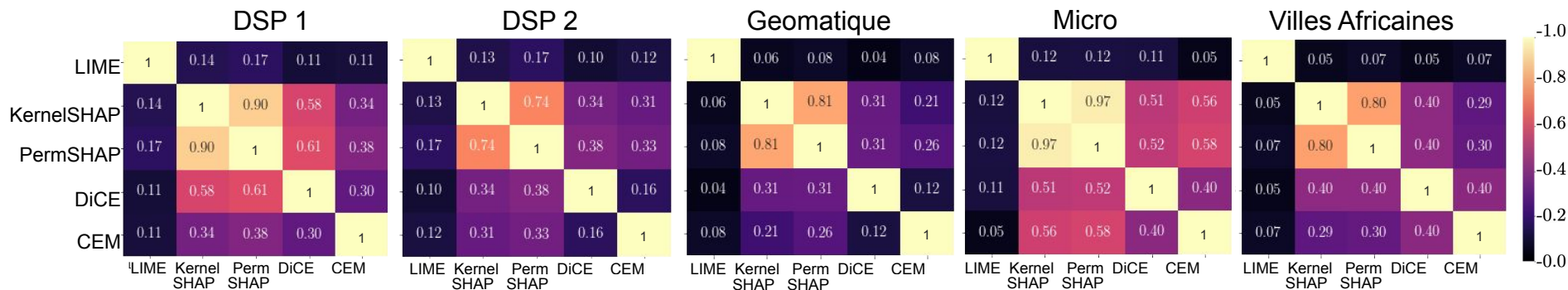
RESULTS



[EDM 2022] Evaluating the Explainers
Swamy, Radmehr, Krco, Marras, Käser

How do explanations (**quantitatively**) compare across courses?

Spearman's Rank Order Correlation



Big differences across explainability methods.

RQ2: 5 Courses

RESULTS

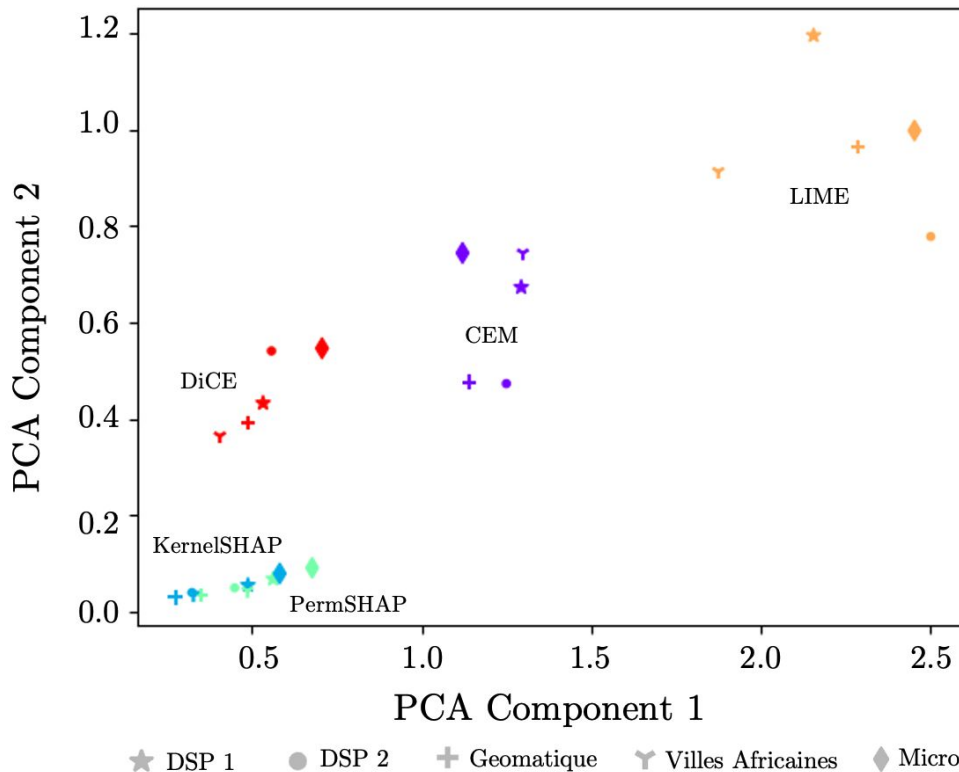
How do explanations
(quantitatively)
compare across
courses?

PCA Analysis

Feature importance clusters
by explainability method,
not by course



[EDM 2022] Evaluating the Explainers
Swamy, Radmehr, Krco, Marras, Käser





**XAI methods
systematically
disagree.**

**How can we build
trust in them?**



Human expert validation!

Study Participants

RESULTS

Who can we most trust to validate educational explainers?



26 STEM professors
45 minute semi-structured interviews

- diverse in geographic location
- 80.95% identify as male
- age: 36.5 yrs (std: 9 years)
- strong MOOC expertise

Expert Trust

RESULTS

- diverse prior perceptions of what factors enable student success and failure
- trust explanations that aligned with their beliefs (46.8%) (which led to significant disagreement across experts)
- only 3 chose the same method for both courses in the pair



Main Takeaways

EVALUATING THE EXPLAINERS

TRUSTING THE EXPLAINERS

Explainability methods, systematically,
do not agree
on which features are important for predictions

.... and neither do human experts



**Where do we go from
here?**

Interpretable-by-design
neural nets!

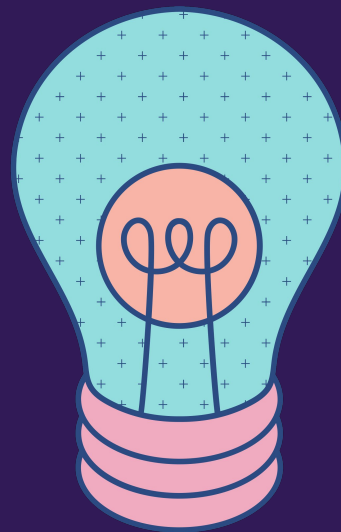
MultiModN

github.com/epfl-iglobalhealth/MultiModN

Multimodal, Multitask, Interpretable Modular Networks



EPFL Yale



Vinitra Swamy^{1*}



Malika Satayeva^{1*}



Jibril Frej¹



Thierry Bossy¹



Thijs Vogels¹



Martin Jaggi¹

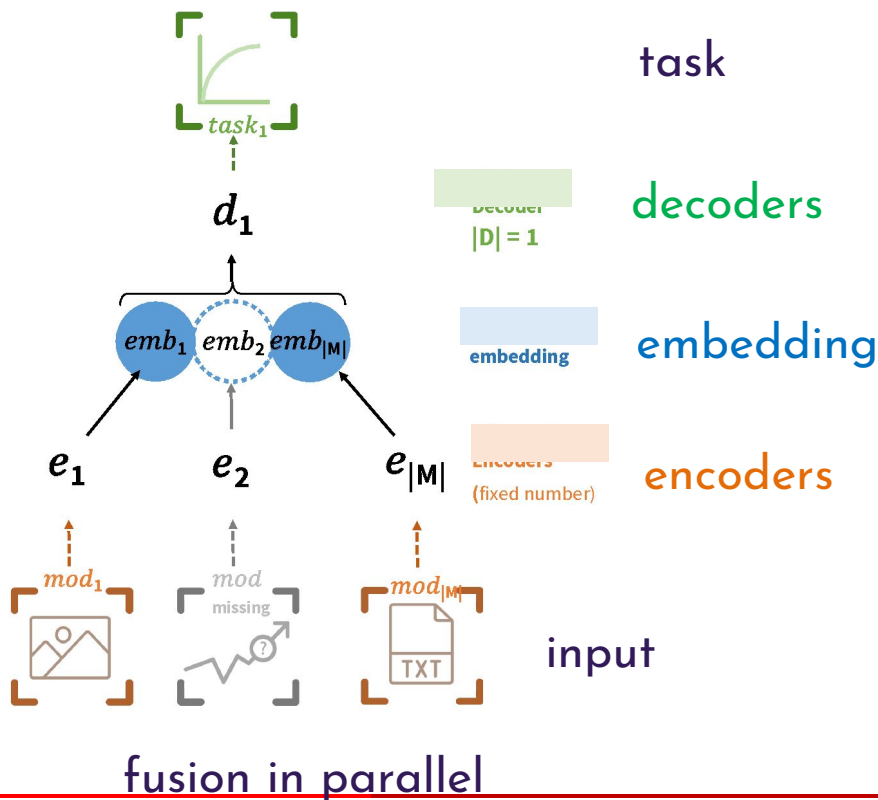


Tanja Käser^{1*}

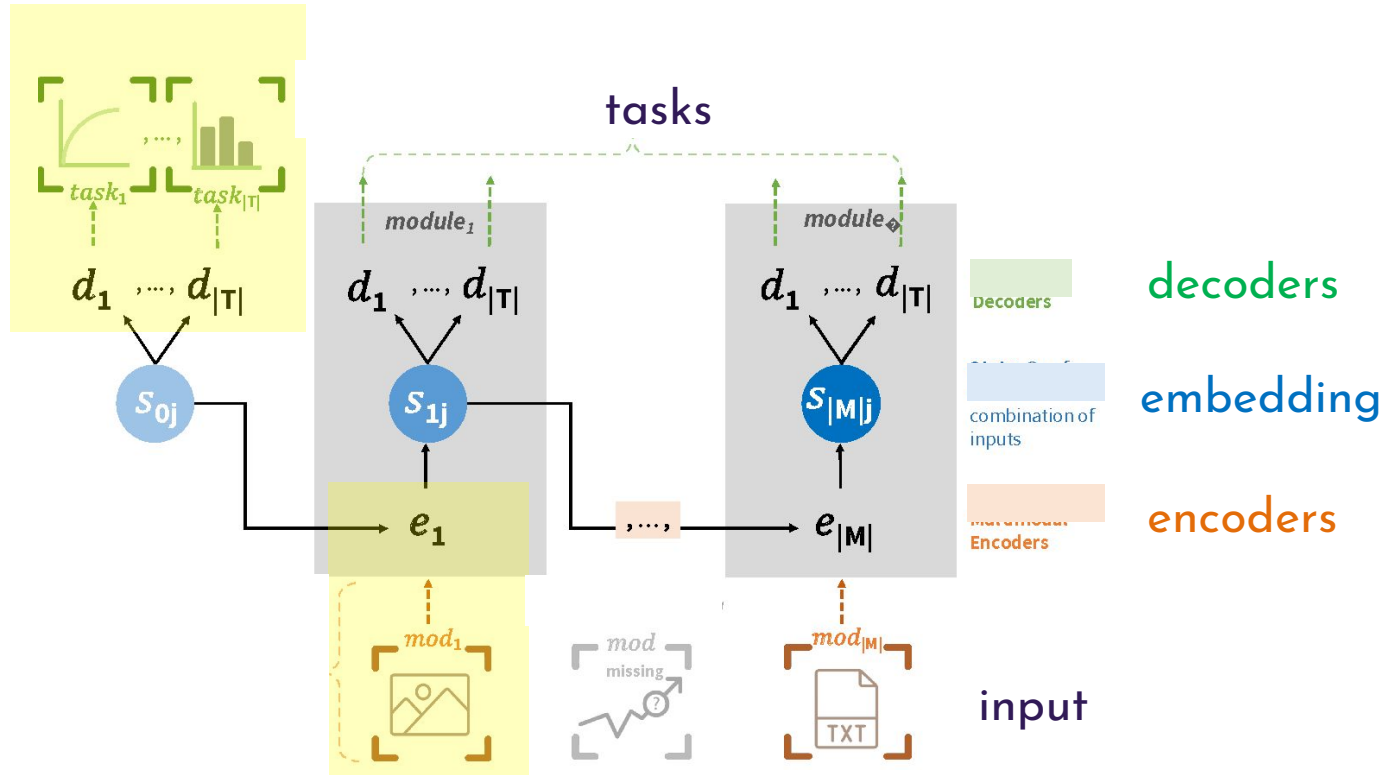


Mary-Anne Hartley^{1,2*}

Traditional Parallel Fusion



MultiModN



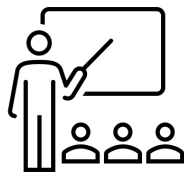
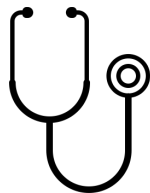
fusion in sequence

Real World Multimodal Settings

MULTIMODN

Evaluated on 10 real world tasks across multimodal
healthcare (MIMIC IV + CXR-JPG), education (EPFL
MOOC), and weather (Weather2k) benchmarks

921 patients in tertiary
care in a Boston hospital



5,611 students in a
MOOC with over
1 million interactions

1866 ground weather
stations covering 6
million km²



✨ From the Lab to Real Life ✨

MULTIMODN



Prof. Annie Hartley
(MD, PhD)



clinical studies using MultiModN for pneumonia and tuberculosis diagnosis in **low resource settings**

- trained more than 100 doctors to collect multimodal data (images, ultrasound)
- recruiting 1000s of patients in South Africa, Tanzania, Namibia and Benin

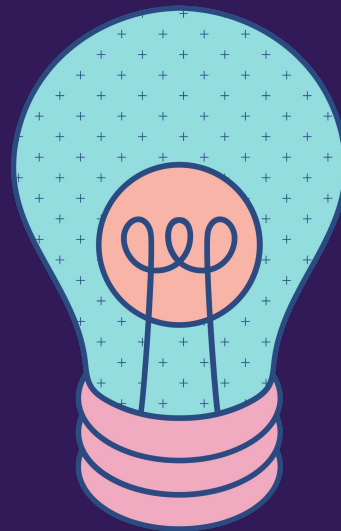
Under
Review

InterpretCC

github.com/epfl-ml4ed/InterpretCC

Intrinsic, User-Centric Interpretability through Global MoE

EPFL



Vinitra Swamy



Syrielle Montariol



Julian Blackwell



Jibril Frej



Martin Jaggi



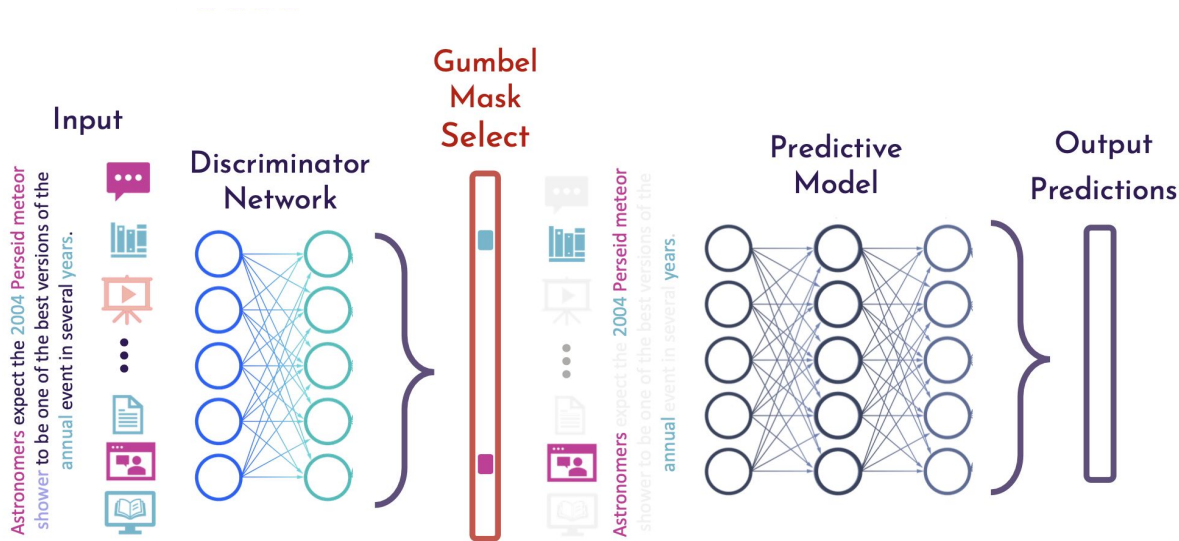
Tanja Käser

InterpretCC: Feature Gating for Interpretability



Swamy, Montariol, Blackwell, Frej,
Jaggi, Käser

Adaptive Feature Gating - accuracy vs. interpretability tradeoff



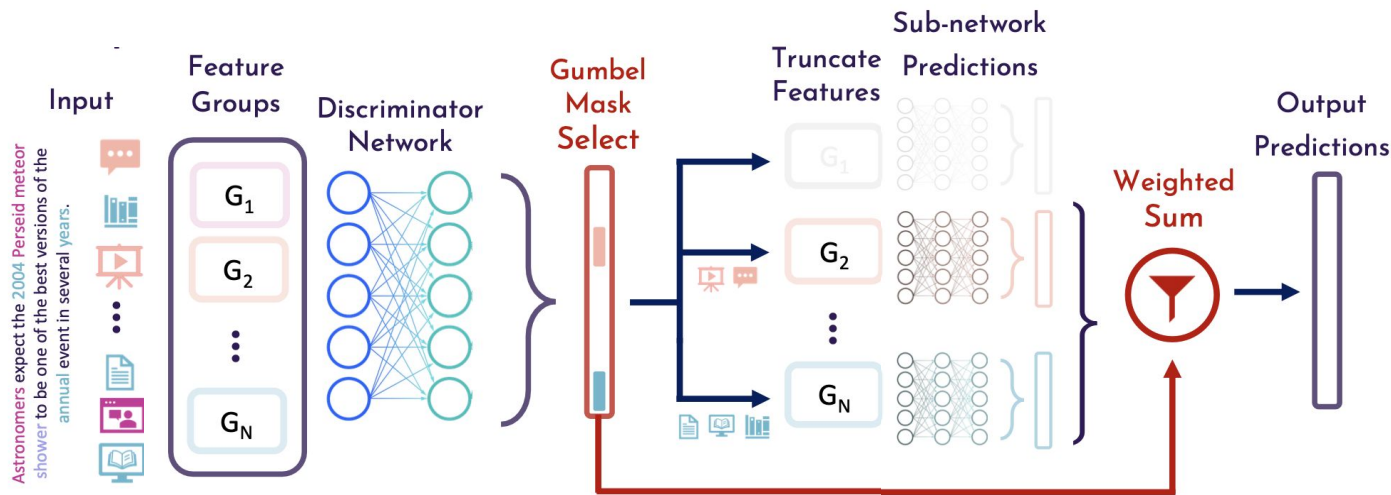
6
7

InterpretCC: Mixture-of-Experts for Interpretability



Swamy, Montariol Blackwell,
Frej,
Jaggi, Käser

Filter the feature space and send relevant parts to relevant experts



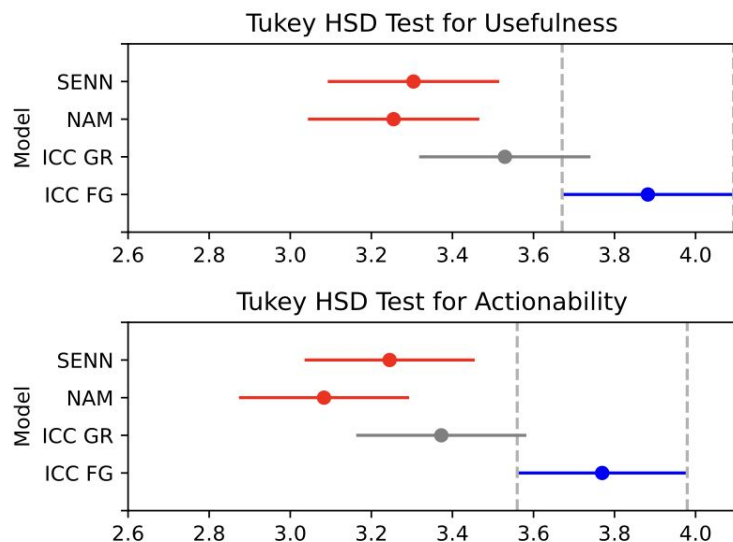
InterpretCC: Results



Swamy, Montariol Blackwell,
Frej,
Jaggi, Käser

Preferred by 56 teachers over other interpretable-by-design approaches

6
9



	NAM	SENN	ICC GR	ICC FG	Weight
Usefulness	3.25 $\pm$ 0.98	3.3 $\pm$ 1.11	3.53 $\pm$ 1.11	3.88 $\pm$ 0.94	0.28
Trustworthiness	3.28 $\pm$ 0.93	3.64 $\pm$ 0.92	3.36 $\pm$ 1.06	3.78 $\pm$ 0.9	0.23
Actionability	3.08 $\pm$ 0.96	3.25 $\pm$ 1.06	3.37 $\pm$ 1.04	3.77 $\pm$ 0.95	0.21
Completeness	3.18 $\pm$ 1.02	3.76 $\pm$ 1.09	3.1 $\pm$ 1.19	3.67 $\pm$ 1.07	0.16
Conciseness	3.13 $\pm$ 1.06	2.82 $\pm$ 1.31	3.72 $\pm$ 1.06	3.68 $\pm$ 1.05	0.12
Global	3.2 $\pm$ 0.81	3.38 $\pm$ 0.85	3.41 $\pm$ 0.88	3.78 $\pm$ 0.77	

Under
Review

iLLuMinaTE

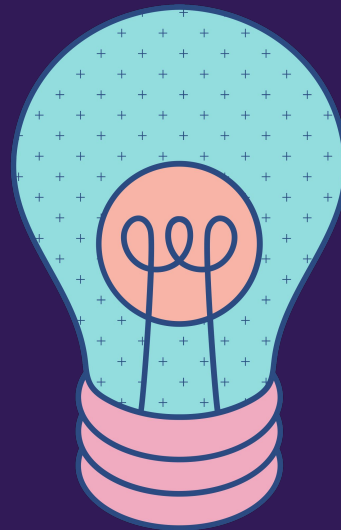
github.com/epfl-ml4ed/illuminate

From Explanations to Action: A Zero-Shot, Theory-Driven
LLM Framework for Student Performance Feedback

EPFL



IP
PR



Vinitra Swamy*



Davide Romano*



Bhargav Desikan



Oana-Maria Camburu



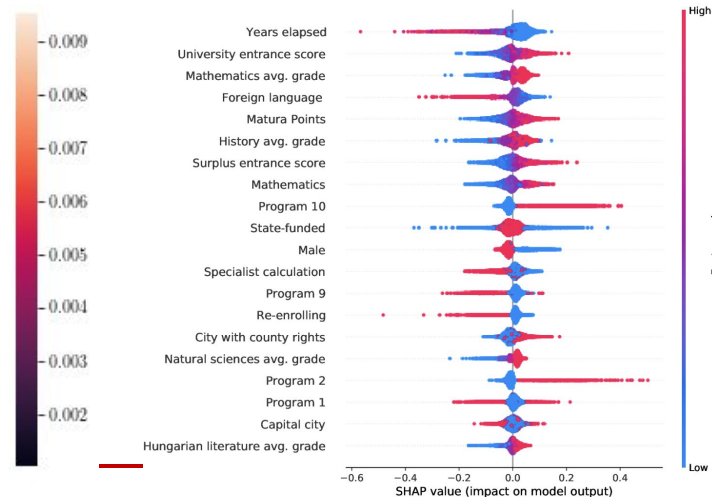
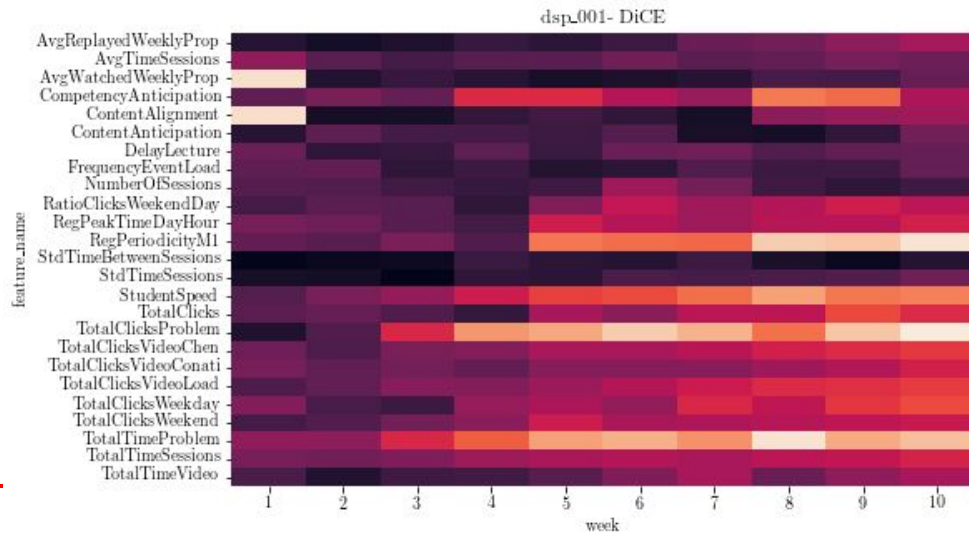
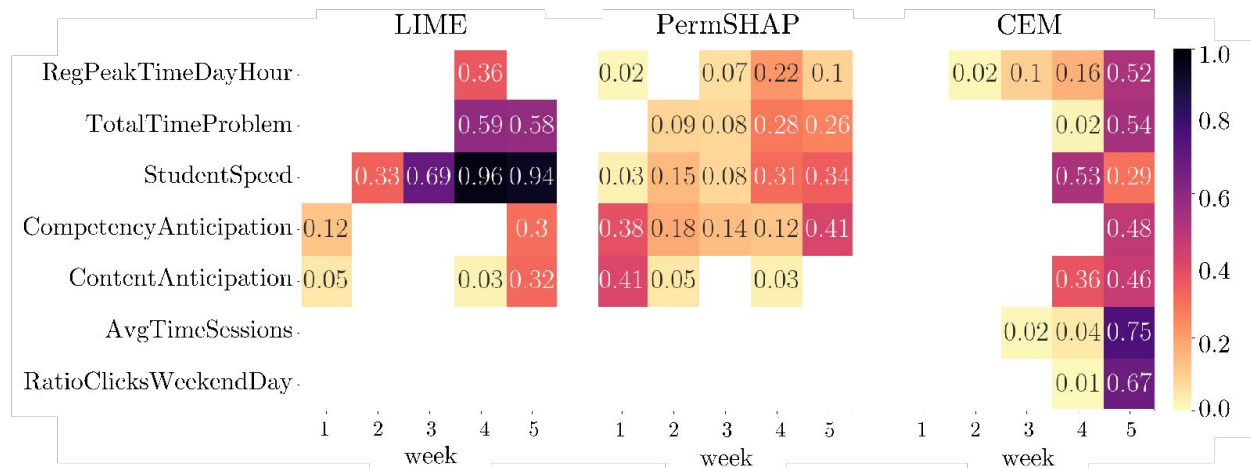
Tanja Käser

Processed Output of an Explainer

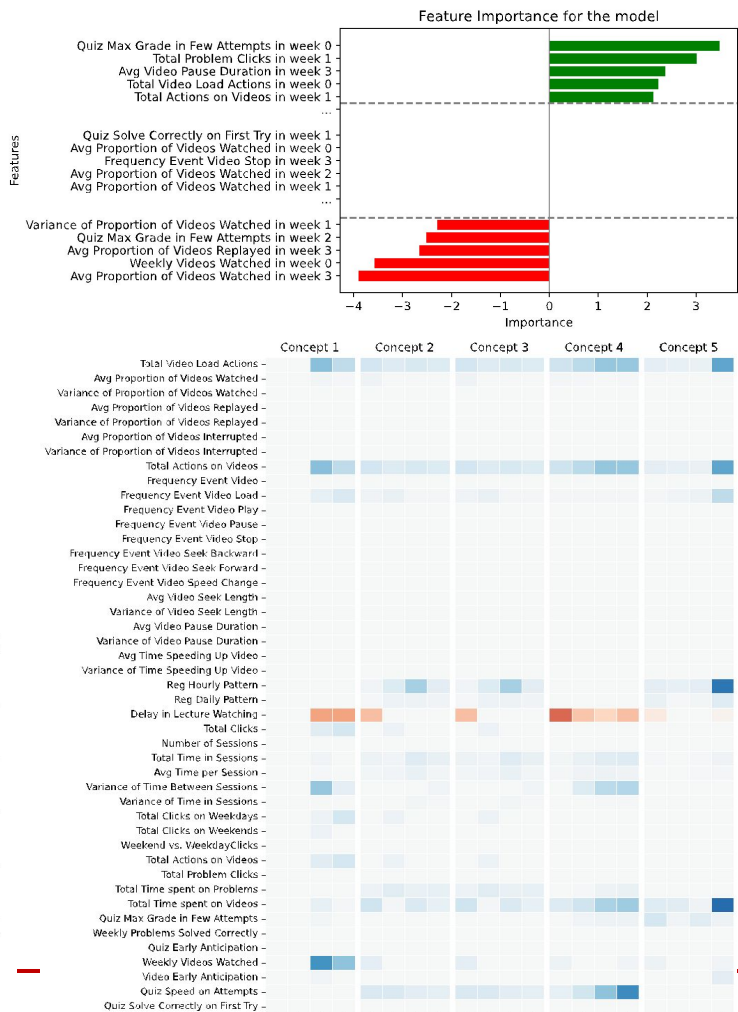
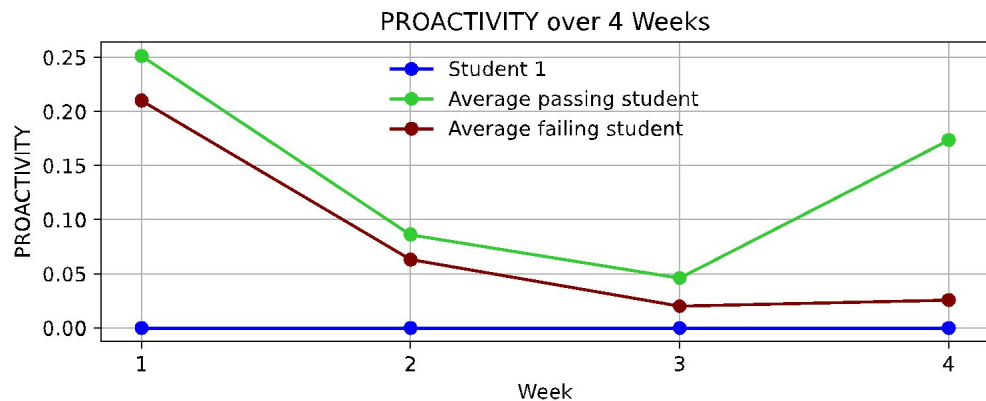
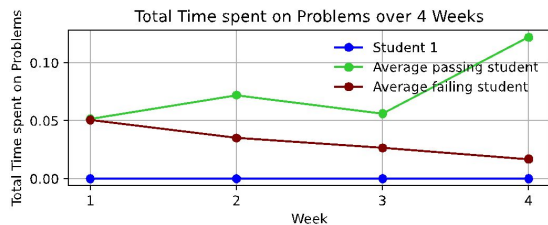
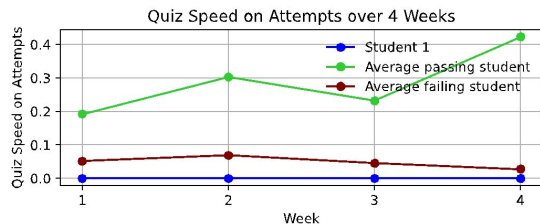
	exp_num	total_clicks_InWeek1	number_sessions_InWeek1	time_in_video_sum_InWeek1	time_in_problem_sum_InWeek1
66	1575	-4.56E-09	-2.21E-10	0.074035813	1.3699654119458948e-09
67	5252	-5.60E-10	-2.21E-10	2.1067425364992842e-11	-2.06E-10
68	881	1.7555813192071668e-09	-8.83E-10	0.14653082042868498	-2.65E-10
69	2683	-3.42E-09	-2.21E-10	0.24476348871729897	1.264484322804904e-09
70	16963	-8.50E-10	0	0.045331784	-1.61E-09
71	5931	6.354141102171695e-10	1.3576282653637861e-08	0.179889493833993513	-7.48E-10
72	12145	0	0	0.9422763586044312	0
73	11999	0	0	0.03497038	0
74	1571	6.909329608451031e-09	-8.83E-10	0.13568544287717593	1.6364350777231529e-09
75	2220	-5.60E-10	0	0.057424471615632494	0
76	9184	0	0	0.025690399	0
77	9592	-2.28E-09	-2.21E-10	0	-1.84E-10
78	12309	5.789162378644352e-10	-2.21E-10	0	-4.92E-10
79	5394	-2.28E-09	0	0.026910097	0
80	730	-2.28E-09	-2.21E-10	0.029592504314488427	8.244599122332608e-13
81	3318	2.315664951457741e-09	6.7881413268189306e-09	0.014542981	-1.48E-09
82	290	-1.12E-09	0	-3.32E-09	0
83	1959	5.789162378644352e-10	0	1.8329978368480937e-09	4.3817753245245505e-10
84	1424	-1.70E-09	-4.42E-10	-4.82E-09	4.4265635254503444e-10
85	12925	0	0	0.021664523	0
86	11139	0	0	0	0
87	12603	0	0	0	0
88	16987	1.1578324757288705e-09	-2.21E-10	0	-1.46E-10
89	10284	0	0	0	0
90	1316	1.7555813192071668e-09	-8.83E-10	0.015449408282339427	-9.00E-10
91	2601	1.447290594661088e-10	0	0	0
92	2851	-2.84E-09	-4.42E-10	0.037342517841102124	0
93	16707	2.894581189322176e-10	0	0	9.635302451738159e-11
94	11108	0	0	0	0
95	5846	1.1578324757288705e-09	-2.21E-10	-3.86E-11	0
96	16538	1.447290594661088e-10	0	0	0
97	14230	0	0	0	0

10 weeks x 45 fts
= 450 columns

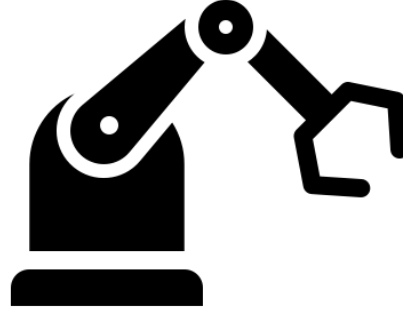
Heavily Processed Visuals



Heavily Processed Visuals



Select what's important

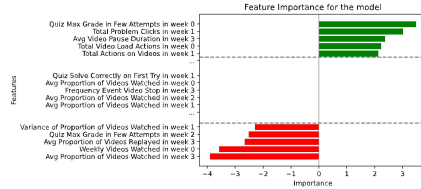
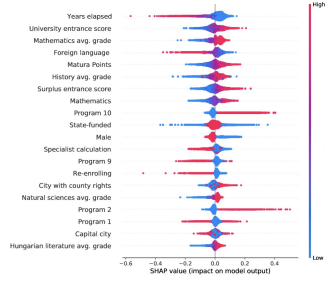


students are happy*!

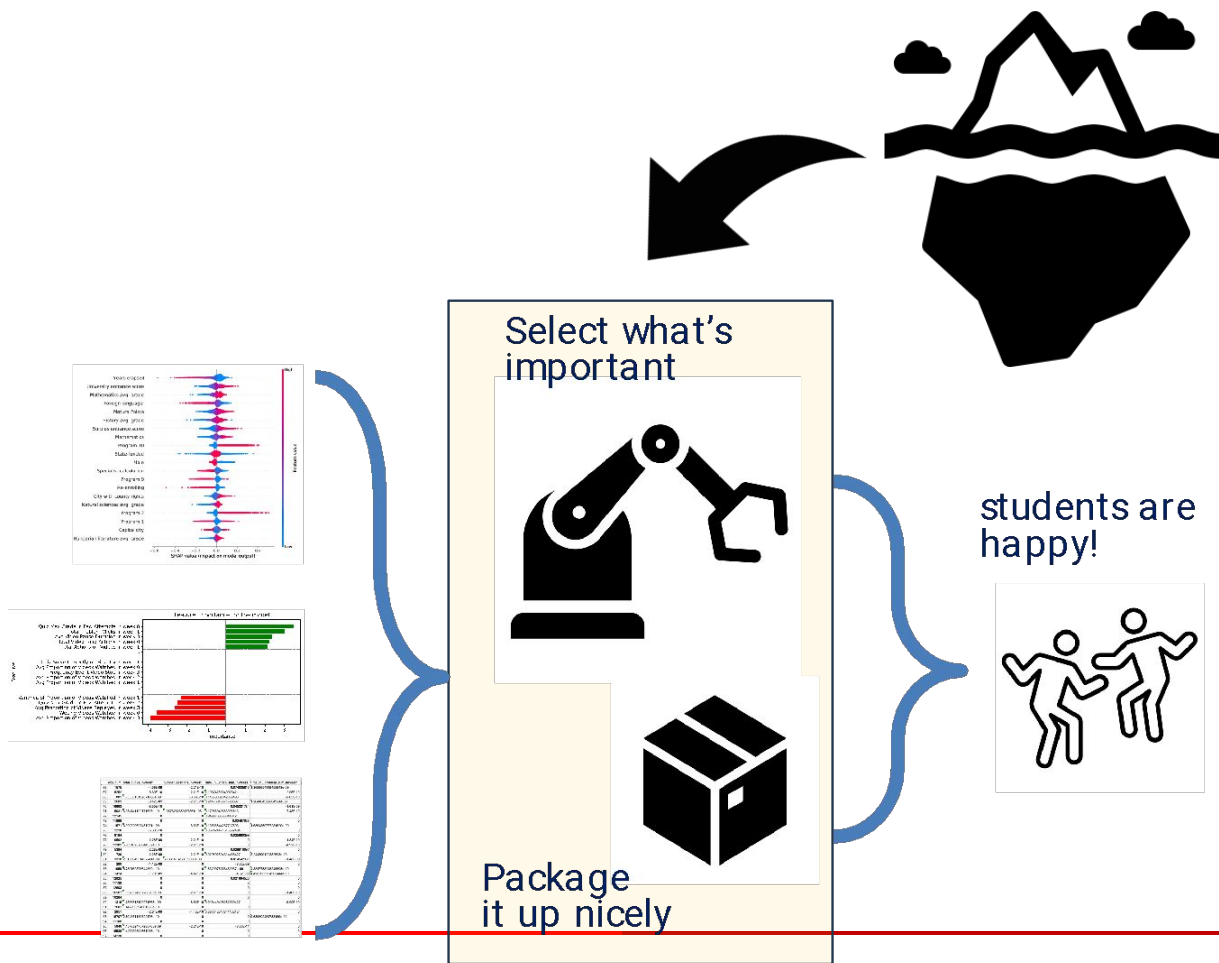


*with their improved learning outcomes

Package it up nicely



id	user_id	course_id	problem_id	score	time	status	feedback
80	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
81	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
82	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
83	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
84	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
85	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
86	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
87	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
88	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
89	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
90	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
91	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
92	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
93	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
94	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
95	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
96	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
97	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
98	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
99	1015	4500-09	2370-10	0.0	0.0	0.0	0.0
100	1015	4500-09	2370-10	0.0	0.0	0.0	0.0



Leverage theories of explanation from philosophy and social sciences to **define prompting strategies**



Hempel
DN model

Relevance
Selection

Statistical
Relevance



C.S. Peirce
Abductive reasoning

Abnormal
Conditions



Malle
Social attribution model

Necessity
Robustness

Base
Contrastive



P. Lipton
Contrastive explanation

RaR +
Contrastive

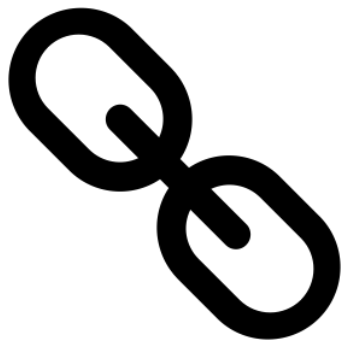


Lombrozo
Cognitive processes of
explanation

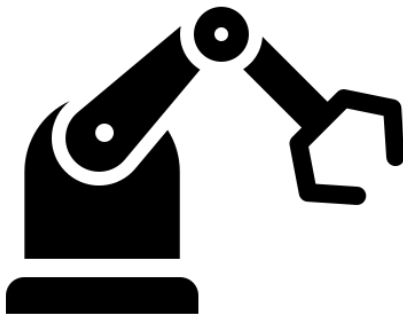
Pearl's
Explanation

Chain-of-Thought

Cognitive Processes of Explanation



Causal
Connection

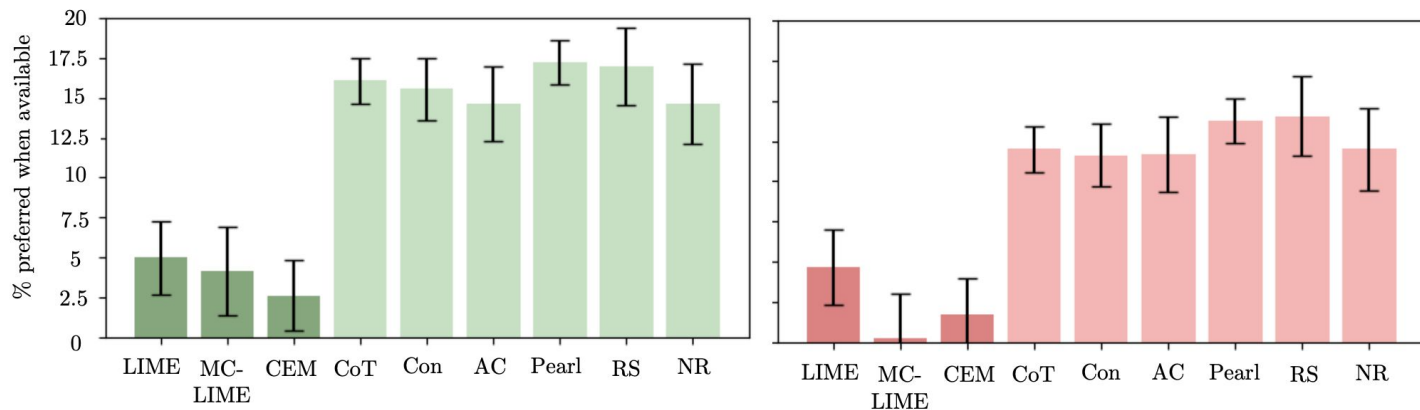


Explanation
Selection



Explanation
Presentation

Results



Students prefer iLLuMinaTE explanations significantly over expert processed post-hoc explanations!

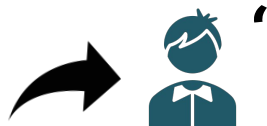
text >> visual part of the explanation

Actionability

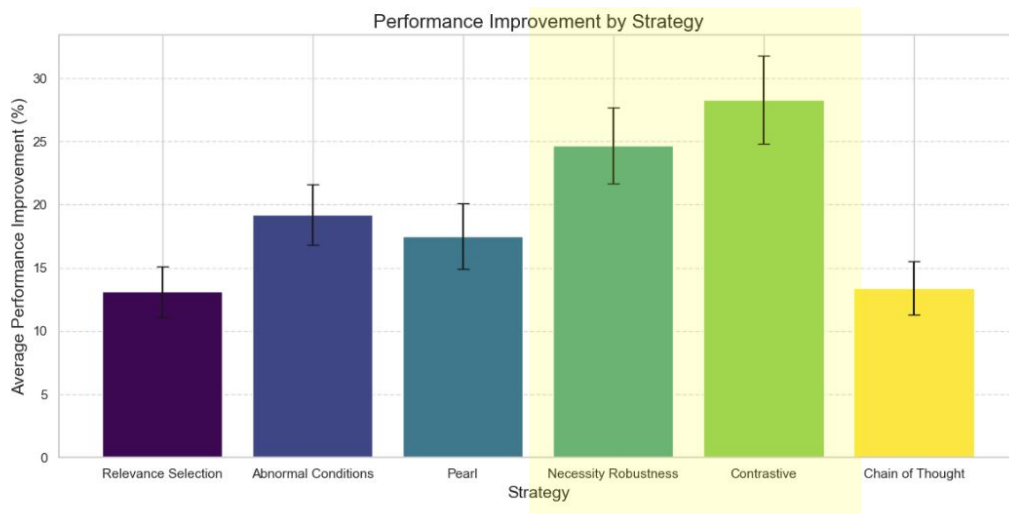
students selected an action to take based on which explanation they prefer



week 5



week 6
simulated
intervention



Contrastive, Necessity Robustness

Summary

- Interpretability is important (not only for education)
 - We can use intrinsic interpretable models or post-hoc methods to get interpretable predictions
 - Methods can be categorized into global and local
 - PDP is easy to interpret, but has an independence assumption and is limited to a low number of features
 - LIME leads to short explanations, but also ignores correlation between features and might lead to instable explanations
-

Agenda

- 1) Introduction to Explainability
 - Taxonomy of interpretability method
 - Deep Dive: PDP
 - Deep Dive: LIME
 - 2) **Personalized Learning with Scholé**
-





Teach same example for everyone → doesn't work perfectly for anyone



Teach same example for everyone → doesn't work perfectly for anyone



Tailor a different example for everyone → tons of human effort, complicated

Curriculum that adapts to you

Scholé **tracks**
each
employee's
progress at the
skill level.



Skill 1



Understand Expert

Skill 2



Apply Beginner

Curriculum that adapts to you

Scholé **tracks**
each
employee's
progress at the
skill level.



Bite-sized
chunk



(Chosen modality)

Skill 1



Understand Expert

Skill 2



Apply Beginner

Curriculum that adapts to you

Schol  tracks
each
employee's
progress at the
skill level.



Bite-sized
chunk

Question
or task

Skill 1



Understand Expert



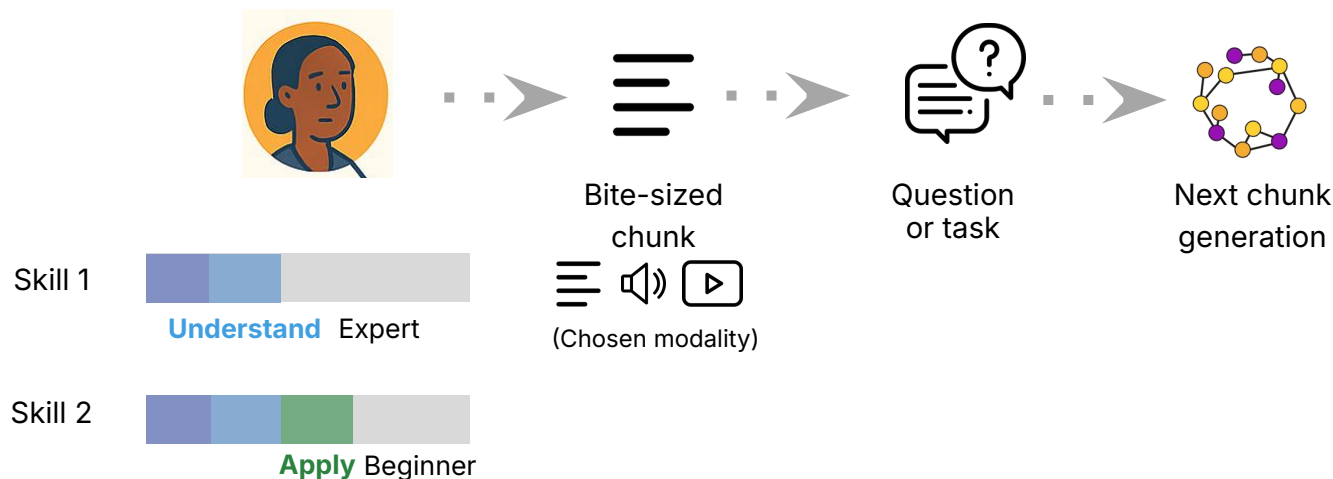
(Chosen modality)

Skill 2



Apply Beginner

Curriculum that adapts to you



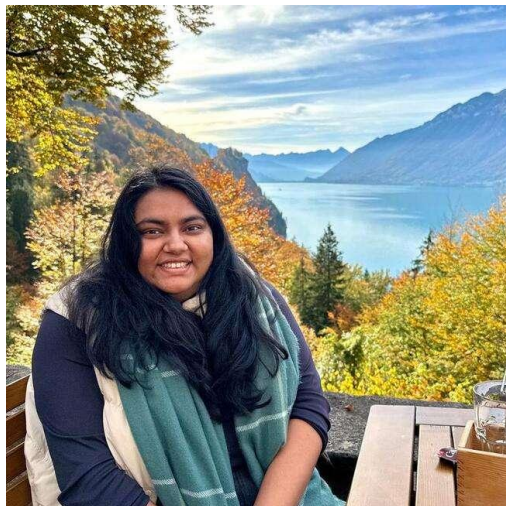
Scholé **tracks**
each
employee's
progress at the
skill level.

The system
dynamically
adjusts the
difficulty of
content, examples
and tasks



<https://app.schole.ai>

"Explainable AI"



Vinitra Swamy

CEO, CS PhD



Paola Mejia

CTO, CS PhD



ML4ED Lab
EPFL



Prof. Tanja Käser
Chief Science Advisor



House Fund
UC Berkeley



Research Students
EPFL



Any Questions?
